Machine Learning

How computers learn from data

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What is Artificial Intelligence?

AI is the ability of machines to do tasks that normally need human intelligence, such as learning, reasoning and decision-making.

1

Narrow AI

Built for one task. All AI today is narrow, e.g. Siri and spam filters.

2

General AI

Matches human ability across any task. Still only theoretical.

3

Super AI

Surpasses human intelligence. Purely hypothetical.

A Short History of AI



AI vs ML vs Deep Learning



Artificial Intelligence

Any technique that makes machines act intelligently.

Machine Learning

Systems that learn patterns from data instead of fixed rules.

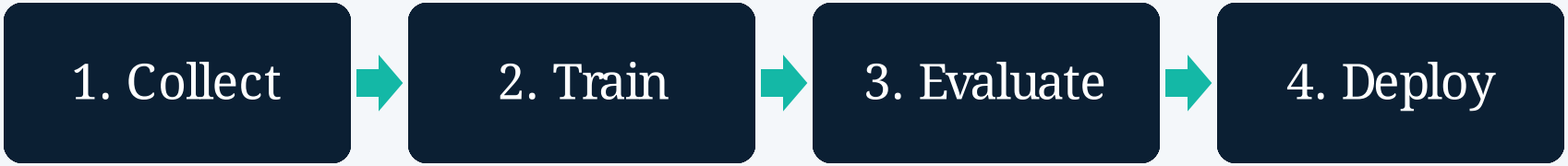
Deep Learning

ML with many-layered neural networks. Powers vision and chatbots.

Three Types of Machine Learning

Supervised	Learns from: Labeled data (inputs with the correct answers) Example: Spam detection, house price prediction
Unsupervised	Learns from: Unlabeled data; it finds hidden patterns itself Example: Customer segmentation, anomaly detection
Reinforcement	Learns from: Trial and error with rewards and penalties Example: Game-playing AI, robotics, self-driving

How a Model Learns



Gather and clean a large, relevant dataset.

The algorithm finds patterns by adjusting its parameters.

Test on unseen data to measure accuracy.

Use the model in a real app and keep monitoring it.

Garbage in, garbage out: data quality decides model quality.

AI in the Real World

H Healthcare

Spots diseases in scans and X-rays

F Finance

Detects fraud and scores credit risk

T Transport

Self-driving cars and route planning

E Education

Personalised learning and tutoring

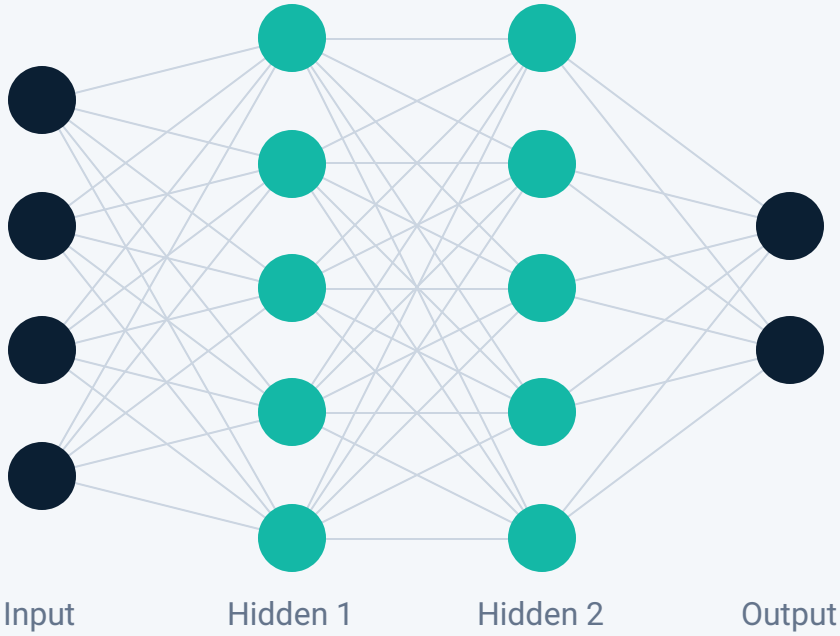
R Retail

Product recommendations and demand forecasts

C Cybersecurity

Flags threats and unusual activity

Neural Networks



- Inspired by the brain: layers of connected nodes (neurons).
- Each connection has a weight that is adjusted during training.
- More hidden layers means a deeper network, which is deep learning.

Challenges & Ethics

1. **Bias** — Models trained on biased data make unfair decisions, e.g. in hiring.
2. **Privacy** — Learning needs lots of data, often personal and sensitive.
3. **Explainability** — Deep models are hard to explain: why did it decide that?
4. **Jobs** — Automation changes the work people do and the skills needed.

Key Takeaways

1. AI is a broad field; machine learning is the part that learns from data.
2. Models are trained on data, tested on new data, then deployed.
3. AI is powerful, but must be built to be fair, private and explainable.

Thank you! Questions?